MIL-W-24126(SHIPS
AMENDMENT-1
1 December 1965

MILITARY SPECIFICATION

WOOD LAMINATES, SOUTHERN PINE (FOR SHIP AND BOAT USE)

This amendment forms a part of Military Specification MIL-W-24126(SHIPS), dated 30 March 1965.

Page 1, paragraph 1.2: Under "Grade" add "Grade AA".

Page 1, paragraph 1.2, class: Delete classes 1 and 2 and substitute the following:

"Class 1 - High decay resistance.
Class 2 - Good decay resistance.
Class 3 - Moderate decay resistance."

Page 1, paragraph 2.1, under Military Specifications: Delete reference to MIL-W-6110 and substitute the following:

"MIL-W-19463 - Wood Moisture Content Determination."

Page 3, paragraph 3.1, line 2: After "specification" add "for the class specified".

Page 3, paragraph 3.1(a), line 1: After "specification" add "for the class ordered,".

Page 3, paragraph 3.2.2, line 5: Delete "(see 2.2)".

Page 5, paragraph 3.2.5.1: Delete and substitute:

"3.2.5.1 Class 1 (high decay resistance).- Lamination shall contain the following:

"(a) Bright sapwood on not less than 75 percent of the area of any face or edge, in any two feet of length, treated in accordance with 3.2.5.1.1 through 3.2.5.1.4; or

"(b) Any combination of heartwood and sapwood, treated as specified in 3.2.5.1 (a), provided that 100 percent penetration of both is achieved."

FSC 5510

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Page 6, paragraph 3.2.5.1.2, last line: Delete "portion" and substitute "or heartwood and sapwood portion (see 3.2.5.1)."

Page 6, paragraph 3.2.5.1.3: Delete and substitute:

"3.2.5.1.3 Treatment.- The treatment shall be any method which does not use a pressure higher than 150 pounds per square inch (p.s.i.) or a temperature higher than 140° Fahrenheit (F.), and provides 100 percent penetration of preservative in the sapwood or heartwood and sapwood with a dry salt retention of not less than 0.50 lbs/ft.^{3}. Incising of the lumber is not permitted."

Page 6, paragraph 3.2.5.2: Delete and substitute:

"3.2.5.2 Class 2 (good decay resistance).- Laminations shall be free of sapwood or may contain unlimited sapwood if treated as specified in 3.2.5.1, with 100 percent penetration of sapwood portion."

Page 6: Add the following new paragraph:

"3.2.5.3 Class 3 (moderate decay resistance).- Laminations may contain bright sapwood on not more than 25 percent of the area of any face or edge, in any two feet of length, or may contain unlimited sapwood if treated as specified in 3.2.5.1 with 100 percent penetration of the sapwood portion."

Page 8, paragraph 3.8.1, line 1: After "After machining," add "class 2 and class 3".

Page 10, table III; columns 1 through 4: Delete "Orientation of annual rings (flat or edge-grain) (see 3.2.3)", "Casehardening (see 3.2)", and "Class (see 3.2.5)" requirements and substitute the following:

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"Orientation of annual rings (flat or edge-grain) (see 3.2.3)	I Laminations or I Lamination components prior to bonding	1.5	a. Number of laminations with predominately the wrong grain exceeds acceptance number.
		4.0	a. Number of pieces with predominately the wrong grain exceeds acceptance number.
Casehardening (see 3.2)	S-3-lamination components prior to end-joining (test 4.5.2)	4.0-case-hardened or oversteamed	a. The number of laminations which are casehardened or oversteamed exceeds the quality acceptance number. (Individual laminations which are casehardened or oversteamed shall be rejected.)
"Class (see 3.2.5) "Class 1	I	2.5 preservative treatment 6.5 heartwood untreated	a. The number of laminations which do not meet the required class requirement, exceeds the acceptance number. (Individual laminations which do not meet the class requirement shall be rejected.) b. If heartwood untreated, any lamination exceeds 50 percent heartwood on any face or edge in any two feet of length.
"Class 2	I	2.5 preservative treated or untreated.	a. The number of laminations which do not meet the required class requirement, exceeds the acceptance number. (Individual laminations which do not meet the class requirement shall be rejected.) b. If untreated, any lamination exceeds 25 percent sapwood, on any face or edge, in any two feet.

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"Class 3	I	2.5 preservative ~~treated~~ or untreated	a. The number of laminations which do not meet the required class requirement, exceeds the acceptance number. (Individual laminations which do not meet the class requirement shall be cause for rejection.) b. If untreated, any lamination exceeds 50 percent sapwood, on any face or edge, in any two feet."

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Page 14, paragraph 4.4.1.2: Delete and substitute:

"4.4.1.2 Examination and test for preservative (treated laminations only).- A 0.2 inch diameter boring shall be taken from each sample board selected in accordance with table III, so that the boring extended through the deepest section of the sapwood into the heartwood or through the heartwood and sapwood as required by the class specified. The borings shall be [illegible] If desired by the laminator, full cross-sections may be substituted for borings for the examinations and tests specified in 4.5.7.1 and 4.5.7.2."

Page 15, paragraph 4.5.1, line 2: Delete "MIL-W-6110" and substitute "MIL-W-19463".

Page 15, paragraph 4.5.4: Delete and substitute:

"4.5.4 Measurements of knots and holes.- Knot or hole sizes shall be determined by averaging the distance between lines drawn parallel and perpendicular to the edges of the piece and tangent to the knot or hole being measured."

Page 17, paragraph 4.5.7: Delete "(class 1)" and substitute "(treated laminations only)".

Page 17, paragraph 4.5.7.1, line 2: Delete "sapwood" and substitute "heartwood and sapwood or heartwood as required by the class specified".

"4.5.7.2 Preservative retention.- According to durability class, the entire heartwood or heartwood and sapwood portion of each boring or cross-section shall be composited and the composite analyzed by any accurate and recognized assay method."

Page 19, paragraph 6.8: Delete and substitute:

"6.8 Laminate class 1/.

"(a) Class 1 (high decay resistance).- For service where high decay or wood destroying insect hazard exists, such as damp, humid, warm areas, having poor ventilation and an exposure to freshwater leakage and condensation, for parts where replacement would be difficult and costly, and long time service is required.

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"(b) Class 2 (good decay resistance).- For service where decay insect hazard is moderately high, or may be high at times, but ventilation and drying is generally possible and long time service is required. May be used under adverse decay conditions where a shorter service life can be tolerated.

"(c) Class 3 (moderate decay resistance).- For service where decay or insect hazard is not high or where occasional wetting occurs but good ventilation prevails and long time service is required; may be used in higher decay hazard locations where replacement or moderate service life can be tolerated; may be used under continuously submerged, water-logged conditions."

Page 19: Add the following footnote:

"1/ Laminate classes do not refer to marine borer resistance. For direct exposure to marine borers (Toredo, Bankia, Limnoria), class 3 laminates, with unlimited, untreated sapwood, should be specified to be treated after all shaping and drilling, in accordance with applicable requirements of TT-W-571."

Preparing activity:
Navy - SH
(Project 5510-N016Sh)

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